

ELECTROMAGNETIC WAVES

Electrostatic $\rightarrow$ Electric field

M.E.C Magnetism $\rightarrow$ Magnetic field

E.M.I $\rightarrow$ Time varying magnetic field produce electric field.

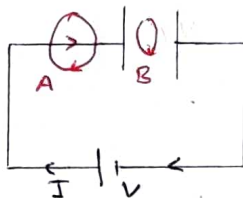
E.M wave $\rightarrow$ Time varying electric field produce magnetic field.

Charge	E.f.	M.f.	EM wave
Rest	✓	✗	✗
$V = \text{const}^n$	✓	✓	✗
Accelerated	✓	✓	✓

Maxwell Experiment

* During charging of capacitor

$\rightarrow$ For the flow of current there must be close & continuous path, but here not a close path.



at 'A' $\rightarrow$ Atq to Ampere-circuital law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{in}, B_A \neq 0 \text{ (theory)}$$

Atq to Maxwell $\rightarrow B_A \neq 0 \text{ (exp)}$

at 'B' $\rightarrow$ Atq, Ampere-circuital law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{in}^0, B_B = 0 \text{ (theory)}$$

Atq to Maxwell $\rightarrow B_B \neq 0 \text{ (exp)}$

Displacement Current

- Formed using changing electric field.
- Proves continuity of current
- Explains existence of magnetic field b/w the plate of capacitor.
- It does not exist under steady state (capacitor full charge) $I_{disp} = 0$.
- Flows b/w cross section area of capacitor plate.

$$I_D = \epsilon_0 \frac{d\Phi_E}{dt} = \epsilon_0 A \frac{dE}{dt} = \frac{\epsilon_0 A}{d} \frac{dV}{dt} = \frac{C}{d} \frac{dV}{dt}$$

$$Q = CV$$

$$I_{dispt} = I_{disp}$$

Four Maxwell's Equation

① Gauss law in electrostatics

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{in}}{\epsilon_0}$$

② Gauss law in magnetism

$\oint \vec{B} \cdot d\vec{a} = 0 \rightarrow$ Monopole does not exist.

③ Induced electric field (Faraday's law)

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}$$

④ Maxwell-Ampere circuital law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 [I_{dispt} + I_{disp}]$$

$$= \mu_0 [I_{dispt} + \epsilon_0 \frac{d\Phi}{dt}]$$

EM wave in LC Oscillation

- Transverse wave
- $Q_{net} = 0$
- Oscillation of electric field & Mag. field $\perp$ to each other
- E. field & Mag field oscillate in same phase

$$B_0 = \frac{E_0}{c} = \frac{E_0}{v}$$

direction of wave

$$\vec{V}_{wave} = \hat{E} \times \hat{B}$$

$\rightarrow$ Place ur four fingers (with palm) of right hand along electric field & slap magnetic field, thumb will indicate direction of wave.

Nature of EMW :-

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

$$V = \frac{1}{\sqrt{\mu_r \epsilon_r}} = \frac{C}{\sqrt{\mu_r \epsilon_r}}$$

$$V = \frac{C}{\mu}$$

$\mu =$ mag. permeability

• Angular frequency

• Angular wave no.

$$\omega = \frac{2\pi}{T} = 2\pi f$$

$$K = \frac{2\pi}{\lambda} = \frac{2\pi v}{C}$$

• Speed of EM wave:-

$$c = \lambda \nu = \frac{\lambda}{T} = \frac{w}{K} = \frac{E_0}{B_0} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

gamma ray	x-ray	ultra-violet	light	IR	micro wave	Radio wave
1 Mev	1 KeV	(10-1000 eV)	(11-2 eV)	10 eV	10 ⁻⁶ eV	10 ⁻⁸ eV
10 ²⁰ Hz	10 ¹⁸ Hz	10 ¹⁶ Hz	10 ¹⁵ Hz	10 ¹² Hz	10 ⁸ Hz	10 ⁴ Hz

$E \downarrow \quad f \downarrow \quad \lambda \uparrow$

Gaddi SUV in My Range

R.W $\rightarrow 1m$ to 10^5m

M.W $\rightarrow 10^{-3}m$ to $1m$

IR $\rightarrow 700nm$ to $10^{-3}m$

VL $\rightarrow 400nm$ to $700nm$

UV $\rightarrow 10^{-8}m$ to $400nm$

x-ray $\rightarrow 10^{-11}m$ to $10^{-8}m$

y-ray $\rightarrow 10^{-14}m$ to $10^{-11}m$

1, 3, 8, 11, 14

Source of EM wave

- ① Accelerating charge
- ② LC-oscillation
- ③ Transition of e^- from n^{th} orbit.
- ④ Retardation of e^- when it enters into a target of high atomic weight (x-ray)
- ⑤ De-excitation of nucleus in radioactivity (y-ray).

* Energy density

• EM wave (U) :-

$$(U) = \epsilon_0 E^2 = \frac{B^2}{\mu_0}$$

$$U_{avg} = \langle U \rangle = \frac{\epsilon_0 E_0^2}{2} = \frac{B_0^2}{2\mu_0}$$

• Electrostatic (U_E) :-

$$U_E = \frac{1}{2} \epsilon_0 E^2$$

$$\langle U_E \rangle_{avg} = \frac{1}{4} \epsilon_0 E_0^2$$

• Magnetic (U_B) :-

$$U_B = \frac{B^2}{2\mu_0}$$

$$\langle U_B \rangle_{avg} = \frac{B_0^2}{4\mu_0}$$

* Intensity of EM wave

$\rightarrow$ Total energy passing through unit Area per-sec.

$$I = \frac{E}{At} = \langle U_{em} \rangle c$$

$$I = U c$$

$$I = \frac{E}{At} = \frac{E}{4\pi r^2 t} = \frac{\text{Power}}{4\pi r^2} \text{ w/m}^2$$

Radiation pressure

$E = Pc \rightarrow$ Momentum

① for a perfectly reflecting surface

$$P = \frac{2I}{c}$$

② for a perfectly absorbing surface

$$P = \frac{I}{c}$$

Poynting vector

Direction:- $\parallel$ to wave

Magnitude:- Intensity of EM wave.

$$\vec{S} = \frac{\vec{E} \times \vec{B}}{2\mu_0}$$